

Test 4 — Answer Key

Each row gives the item number, the B.E.S.T. benchmark and difficulty, the correct answer, and a student-facing explanation.

#	Benchmark	Answer	Explanation
1	MA.6.NSO.4.1 easy	-5.0	Combine the two integers, then find that value on the number line.
2	MA.6.NSO.1.3 moderate	20.3	Absolute value is the distance from 0, so it is never negative.
3	MA.6.GR.2.1 challenging	6	Work backwards: double the area, then divide by the height.
4	MA.6.GR.1.3 challenging	18	Find the side lengths by subtracting coordinates, then apply the formula.
5	MA.6.AR.1.2 challenging	$x \geq 60$	A closed circle includes the endpoint; an open circle does not. The shading shows which direction makes the inequality true.
6	MA.6.AR.1.1 challenging	a number x decreased by 8	Read the expression one operation at a time and match it to the words.
7	MA.6.NSO.1.1 moderate	-3.5	On a number line the least number is the one farthest to the left.
8	MA.6.DP.1.3 easy	Minimum $\rightarrow$ 6; Median $\rightarrow$ 15; Upper quartile $\rightarrow$ 18	Read the five-number summary from left to right on the box plot.
9	MA.6.AR.2.4 easy	5.23, 8.77	Undo the operation shown in each equation.
10	MA.6.GR.2.3 easy	2 by 3 by 4 inches $\rightarrow$ 24; 2 by 5 by 6 inches $\rightarrow$ 60; 3 by 3 by 5 inches $\rightarrow$ 45	Multiply the three dimensions of each prism.
11	MA.6.AR.2.2 easy	11, 38	Undo addition with subtraction, and undo subtraction with addition.
12	MA.6.GR.2.4 easy	60, 36	Each pair of opposite faces has the same area, so double the area of one.
13	MA.6.AR.3.3 easy	6, 15	Multiply both numbers in a ratio by the same factor to keep it equivalent.
14	MA.6.AR.3.1 easy	8, 5	Divide both quantities by their greatest common factor.
15	MA.6.AR.3.4 easy	48, 120	Write each percent as a fraction of 100 and multiply by the total.
16	MA.6.NSO.1.4 easy	-35.0, 35.0	Depths of -35 and 35 feet are both 35 units from 0.
17	MA.6.NSO.3.5 easy	$\frac{1}{4} \rightarrow 25\%$; $0.5 \rightarrow 50\%$; $\frac{7}{10} \rightarrow 70\%$	Write each number as a decimal and multiply by 100 to get its percent.

#	Benchmark	Answer	Explanation
18	MA.6.GR.1.1 challenging	(3, -5)	Reflecting across the y-axis changes the sign of the x-coordinate only.
19	MA.6.NSO.1.2 moderate	-14	Below the surface is negative: $-25 + 11 = -14$.
20	MA.6.NSO.3.2 easy	5, 5, 7	Divide each addend by the GCF: $25 \div 5 = 5$ and $35 \div 5 = 7$.
21	MA.6.NSO.3.3 easy	$5^2 \rightarrow 25$; $5^3 \rightarrow 125$; $2^2 \rightarrow 4$	The exponent tells how many times the base is used as a factor.
22	MA.6.AR.1.3 challenging	4	Work inside the parentheses first, then divide.
23	MA.6.DP.1.4 easy	17	Count every dot on the plot, including repeats.
24	MA.6.NSO.1.4 moderate	-63	Farthest from sea level means the greatest absolute value.
25	MA.6.NSO.2.3 easy	\$32.95, \$6.59	Subtract the cost of the pass, then divide what is left by the number of people.
26	MA.6.NSO.2.1 easy	6.75, 108	Multiplying by a number less than 1 makes the result smaller; dividing by it makes the result larger.
27	MA.6.DP.1.5 easy	1	Count the values from the list that fall inside that interval.
28	MA.6.AR.2.1 easy	15 $\rightarrow$ False; 18 $\rightarrow$ False; 20 $\rightarrow$ True; 29 $\rightarrow$ True	Values of 20 or greater make the inequality true.
29	MA.6.NSO.4.2 easy	56, -7	Same signs give a positive result; different signs give a negative result.
30	MA.6.NSO.2.2 easy	$5/6 \times 4/9 \rightarrow 10/27$; $5/6 \div 4/9 \rightarrow 15/8$; $4/9 \div 5/6 \rightarrow 8/15$	Multiply straight across; to divide, multiply by the reciprocal.
31	MA.6.NSO.3.1 easy	GCF of 12 and 18 $\rightarrow$ 6; LCM of 12 and 18 $\rightarrow$ 36; GCF of 12 and 36 $\rightarrow$ 12	The GCF is the largest shared factor; the LCM is the smallest shared multiple.
32	MA.6.AR.3.5 easy	2 pounds, in ounces $\rightarrow$ 32; 3 gallons, in quarts $\rightarrow$ 12; 5 meters, in centimeters $\rightarrow$ 500	Multiply by the number of smaller units in one larger unit.
33	MA.6.DP.1.6 easy	Mean $\rightarrow$ $18 \frac{1}{3}$; Median $\rightarrow$ $14 \frac{1}{2}$; Range $\rightarrow$ 32	Recompute each measure using the six values of the new data set.

#	Benchmark	Answer	Explanation
34	MA.6.DP.1.1 easy	How many text messages did each student send yesterday?, What are the heights, in inches, of the students in the band?	Look for questions that many people would answer with different numbers.
35	MA.6.NSO.3.4 easy	2, 5	A prime factor divides the number and has exactly two factors.
36	MA.6.DP.1.2 easy	Mean → 18; Median → 18 1/2; Range → 12	Find the mean by dividing the total, the median as the middle value, and the range as the greatest minus the least.
37	MA.6.AR.3.2 easy	16 miles in 4 hours → 4; 24 dollars for 2 pounds → 12; 54 pages in 6 minutes → 9	Divide the first quantity by the second in each rate.
38	MA.6.AR.2.3 easy	9, 18	Divide to undo multiplication; multiply to undo division.